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Integrated solar combined cycle system with steam methane reforming: Thermodynamic analysis

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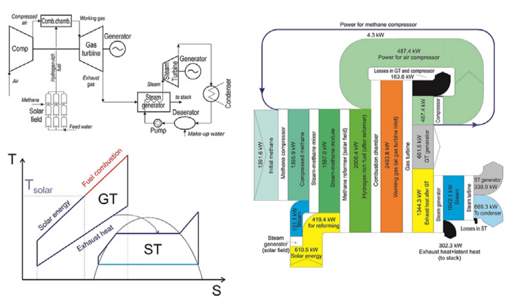
HIGHLIGHTS

- Integrated solar combined cycle system (ISCCS) with steam methane reforming (SMR).
- Efficiency of ISCCS is affected temperature, pressure, and steam-to-methane ratio (β).
- Efficiency of ISCCS with SMR is up to 6.2% and 8.9% for $\beta = 1.0$ and $\beta = 2.0$ higher than it for a conventional ISCCS.
- The share of power output in ISCCS with SMR: 2/3 in gas turbine, and 1/3 in steam turbine.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 4 January 2023
Accepted 23 January 2023
Available online 15 February 2023

Keywords:
Steam methane reforming
Gas turbine
Combined cycle
Thermodynamic analysis

ABSTRACT

The present paper considers an integrated solar combined cycle system (ISCCS) with an utilization of solar energy for steam methane reforming. The overall efficiency was compared with the efficiency of an integrated solar combined cycle system with the utilization of solar energy for steam generation for a steam turbine cycle. Utilization of solar energy for steam methane reforming gives the increase in an overall efficiency up to 3.5%. If water that used for steam methane reforming will be condensed from the exhaust gases, the overall efficiency of ISCCS with steam methane reforming will increase up to 6.2% and 8.9% for $\beta = 1.0$ and $\beta = 2.0$, respectively, in comparison with ISCCS where solar energy is utilized for generation of steam in steam turbine cycle. The Sankey diagrams were compiled based on the energy balance. Utilization of solar energy for steam methane reforming increases the share of power of a gas turbine cycle: two-thirds are in a gas turbine cycle, and one-third is in a steam turbine cycle. In parallel, if solar energy is used for steam generation for a steam turbine cycle, than the shares of power from a gas and steam turbine are almost equal.

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https://doi.org/10.1016/j.ijhydene.2023.01.284
0360-3199/

Introduction

A transition to a carbon-free economy is the reality of the modern energy industry. Reduction in CO_2 emission is one of the main challenge in energy engineering in the last decades. Renewable energy sources are playing an important role on the way to a zero-carbon economy [1,2]. Solar energy is one of the main and almost unlimited energy sources in the World. The different technologies of solar energy use have been developed in the last years [3–8]. However, even though the progress in the development of solar energy technologies is notable, there are a lot of challenges for energy science. One of them is the fact that more than 60% of electricity is produced by conventional technologies via hydrocarbon fuel combustion: steam turbines, gas turbines, etc. While the share of electricity produced by using solar energy is no more than a few percent [9].

Among various ways of utilization of solar energy for electricity generation, a combination of solar energy with the traditional steam and gas turbine cycles can be highlighted. The power plants where solar energy is combined with conventional power cycles are named integrated solar combined cycle systems (ISCCS). In these systems, solar energy is used to produce heat and after that heat is used to generate mechanical work or electricity.

Combined cycle power plants (CCPP) show one of the highest energy efficiency among conventional power plants [10]. The modern cycles with high-temperature gas turbines have an efficiency up to 70% and even higher. In such cycles, the high-temperature gas turbines with the turbine inlet temperature (TIT) up to 1600 °C are applied [11,12]. In the last years, a lot of various integrated solar combined cycle systems (ISCCS) were developed by various scientists and engineers. The main way to use solar energy in such cycles is a steam generation in CCPP [13–16]. In other words, solar energy in such ISCCS is utilized as an energy source in a steam turbine cycle.

The solar fields allow to achieve the high-grade temperature up to 1000 °C [17–19]. Usually, the steam cycles in CCPP are operating under the temperatures up to 600–650 °C. Thus, the high-grade energy source for the steam cycle is difficult to apply. One of the ways to use high-temperature energy is air preheating after a compressor in a gas turbine cycle [20,21]. In this case, the compressed air can be heated by solar energy up to 1000 °C. However, air heating after the compressor is quite a challenging issue because air temperature after the compressor is quite high and additional heating will have low efficiency due to a low-temperature drop. Another way for the integrated use of solar energy is fuel heating before combustion [22]. However, this way doesn't allow for the utilization of a large amount of solar energy because the mass flow rate of fuel is notable less than the mass flow rate of air, and the high temperature of hydrocarbon fuel is the reason for fuel decomposition with free carbon formation. Therefore, hydrocarbon fuel heating is not a widely used way for solar energy utilization in ISCCS.

One of the promising way to use solar energy on a fuel cycle side is a fuel reforming before combustion. In this case, solar energy is used for the endothermic reactions of fuel reforming. Fuel reforming before combustion is widely discussed way of energy efficiency improving. If exhaust heat is used for fuel reforming before combustion, than this way of exhaust heat recuperation is called as thermochemical recuperation [23–26]. In thermochemical recuperation systems, waste heat is utilized for fuel reforming to obtain synthesis gas and then this synthesis gas is used as a fuel in fuel-consuming equipment. Carapellucci and co-workers proposed to use thermochemical recuperation for the gas turbines [27]. Sadeghi an co-authors [28] calculated the efficiency of gas turbine with thermochemical recuperation of exhaust heat via steam methane reforming and showed that the efficiency of such gas turbines can be increased up to 27.52%. Lui and co-authors showed the results of investigation of the combustion performances of the gas turbines with thermochemical recuperation [29] and concluded that the gas turbines with thermochemical recuperation of exhaust heat may be applied for the reduction in NO_x emission.

Solar energy utilization in the steam methane reforming process is widely used technique in the energy storage systems [30–32]. The chemical reactions of SMR process are endothermic and accompanying by water gas-shift (WGS) reaction may be presented as the following:

$$CH_4 + H_2O \Leftrightarrow CO + 3H_2, \Delta H_{298} = 206.1 kJ/mol, \quad (1)$$

$$CH_4 + 2H_2O \Leftrightarrow CO_2 + 4H_2, \Delta H_{298} = 165.0 kJ/mol, \quad (2)$$

$$CO + H_2O \Leftrightarrow CO_2 + H_2, \Delta H_{298} = -41.1 kJ/mol. \quad (3)$$

One of the main advantage of such technique is a potential for transformation of exhaust heat into chemical energy of new synthetic fuel (reforming products). There is a set of publications on the use of solar energy for methane reforming the gas turbine units.

Bianchini and co-workers [33] suggested utilization of solar energy in the steam methane reforming process to obtain synthesis gas and then to use this new synthesis gas as a fuel for the gas turbine. Authors reported that such way of the use of solar energy gives a significant potential in the fuel efficiency increasing they reported that methane saving in such systems is up to 20%.

Lui and co-authors [34] reported the results of the design, modeling, construction, and testing of 100 kW power generation plant based on an internal combustion engine with a solar thermochemical process. In their study, ICE fired with the products of the methanol decomposition process. Solar energy is used as a heat source for this process. The authors reported that the solar-ICE system gives up to 16.71% in the efficiency improving comparing with the systems where methanol is directly burned.

The above mentioned examples depict that solar energy use for methane reforming is quite interesting way of energy efficiency improving in the power plants. Therefore, a detailed analysis of systems should be provided as the first step for the further development of new concepts. Most of the works mentioned above are considering solar energy as free energy. However, solar energy itself can be used for electricity generation, e.g. in the steam turbine cycles. The steam temperature in the steam turbine cycles is no more than 600–650 °C. In such case, the efficiency of solar energy use is limited by the steam temperature. In the gas turbine cycles, the gas temperature can be reached up to 1500–1600 °C. Thus, solar energy use in the gas turbine cycles leads to more efficient utilization, because the thermal efficiency in a gas turbine cycle is usually higher than a thermal efficiency in a steam turbine cycle. Solar energy can be used to generate synthesis gas (after methane reforming) with temperature up to 1000 °C, and then this syngas can be utilized for combustion. Moreover, an efficiency of SMR significantly depends on the technological variables, for example, temperature, pressure, and water-to-fuel ratio.

In this paper, an integrated solar combined cycle system (ISCCS) with the utilization of solar energy in steam methane reforming before combustion is proposed. The thermodynamic analysis of ISCCS where solar energy is used for steam methane reforming is performed to determine an effect of technological variables on a power plant efficiency. To show the benefits of the proposed combined system, the energy efficiency is compared with the conventional integrated solar combined cycle system with the use of solar energy for steam generation for the steam turbine cycle.

ISCCS concept

To understand the main concept of the integrated solar combined cycle system (ICSSC) with steam methane reforming, two schematic diagrams are presented in Fig. 1. The first schematic diagram is for ISCCS with an utilization of solar energy in steam methane reforming (Fig. 1a). The second one is a conventional ISCCS with the use of solar energy for steam generation for the steam turbines (Fig. 1b). A key difference in these diagrams is solar energy use. In the scheme on Fig. 1a, solar energy is utilized in the gas turbine unit; in the scheme on Fig. 1b, solar energy is utilized in a steam turbine unit. Usually, the steam for a steam turbine cycle is limited by the temperature range of 550–650 °C [35]. However, solar energy as a heat source can be used to produce hydrogen-rich gas up to 1000 °C or even higher for a gas turbine [36,37].

In the schematic diagram in Fig. 1a, a solar field for solar energy harvesting is divided into two zones. The first zone is a steam generator to produce steam for steam methane reforming. The second zone is the steam methane reformer where the chemical reactions are taking place.

In the scheme in Fig. 1a, the products of methane reforming are mainly consisting of hydrogen, carbon monoxide, and steam (for the case when a steam-to-methane ratio is higher than 1.0). Also, a small mole fraction of CO_2 produced by water-gas shift (3) reaction and unreacted methane is presented in reformate. The chemical reactions to produce these elements are presented above (1)–(3). These three reaction are the main reactions in SMR process [38]. However, other reaction can take place in the reaction space

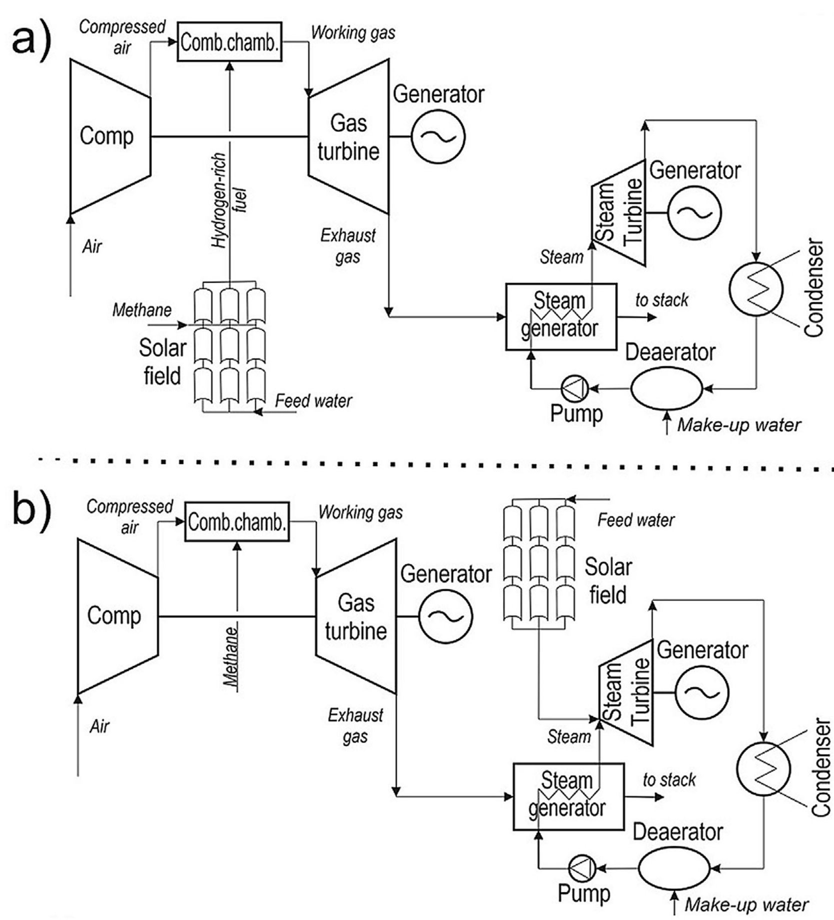

Fig. 1 – The schematic diagrams of an integrated solar combined cycle system with use of solar energy for steam methane reforming (a) and a conventional integrated solar combined cycle system with use of solar energy for steam generation (b).

of the real reformer such as carbon formation reactions. There are sufficient results in the literature that carbon deposition and formation during steam methane reforming can be controlled by technological parameters such as water-to-carbon ratio and temperature [39]. After steam methane reforming, synthesis gas utilizes as fuel in the combustion chamber of the gas turbine.

To better clarify the advantages of a system where solar energy is fed for steam methane reforming, the thermodynamic cycles are presented in Fig. 2. Fig. 2a shows the thermodynamic cycle of an integrated solar combined cycle system with using solar energy for steam methane reforming. Fig. 2b depicts a conventional integrated solar combined cycle system with using solar energy for steam generation for a steam turbine.

In the cycle from Fig. 2a, the heat for the gas turbine cycle is supplied by combustion of the products of reforming. However, part of the chemical energy of hydrogen-rich fuel was obtained via the transformation of solar energy into chemical energy. Moreover, hydrogen-rich fuel has a temperature up to 1000 °C and the enthalpy of this flow is also high. Therefore, in the cycle in Fig. 2a, the heat sources for the gas turbine cycle is virtually divided into two part: solar energy and fuel combustion (chemical energy of initial methane). The thermodynamic cycle for a conventional ISCCS with using solar energy for steam generation in Fig. 2b shows that solar energy and exhaust heat is used to generate steam for a steam turbine cycle.

In addition, Fig. 2 shows that solar energy in the scheme (a) is used with a higher temperature than solar energy for the scheme (b). However, the efficiency of the steam methane reforming process is strongly dependent on operational parameters such as temperature, pressure, and steam-to-methane ratio. The previous studies show that for efficient reforming of methane the temperature of the process has to be higher than 700–800 K [40,41]. Therefore, an effect of technological variables on an overall efficiency of an integrated solar combined cycle system where solar energy is used for steam methane reforming was investigated n this paper. And the obtained results were compared with the overall efficiency of the conventional integrated solar combined cycle system using solar energy for steam generation in a steam turbine cycle.

Methodology

CCPP analysis

An overall efficiency of CCPP is a main criterion of effectiveness of such plants. A thermal efficiency of an integrated solar combined cycle system (ISCCS) with steam methane reforming can be determined as follows:

$$\eta_{ISCCS} = \frac{N_{ISCCS}}{\dot{m}_{CH_4} \cdot LHV_{CH_4}} \tag{4}$$

where N_{ISCCS} is output power of ISCCS with steam methane reforming; $\dot{m}_{CH_4}$ is CH_4 mass flow rate; LHV_{CH_4} is lower heating value of CH_4.

The cycle of ISCCS consists of two subsystems: gas and steam turbine cycles. Thus, an overall output power in a combined cycle may be determined based on an energy balance of the system:

$$N_{ISCCS} = (N_{GT} - N_{a.comp} - N_{m.comp}) + (N_{ST} - N_{pump}) \tag{5}$$

where N_{GT}, N_{ST} – gas and steam turbine power, respectively; $N_{a.comp}$, $N_{m.comp}$ – air and methane compressor power, respectively; N_{pump} – pump power.

A key difference between an ISCCS where solar energy is utilized for steam methane reforming and a traditional ISCCS where solar energy is utilized for steam generation is a pathway of the use of solar energy. In ISCCS with steam methane reforming, solar energy is utilized in a gas turbine cycle through fuel reforming before combustion; in a traditional ISCCS solar energy is utilized in a steam turbine cycle. Thus, an efficiency of gas and steam cycles is calculated as the following [42]:

$$a)\ \eta_{GT} = \frac{N_{GT}}{\dot{m}_{CH_4} \cdot LHV_{CH_4} + Q_{solar}}; \quad b)\ \eta_{GT} = \frac{N_{GT}}{\dot{m}_{CH_4} \cdot LHV_{CH_4}}; \tag{6}$$

$$a)\ \eta_{ST} = \frac{N_{GT}}{Q_{exh}}; \quad b)\ \eta_{ST} = \frac{N_{GT}}{Q_{exh} + Q_{solar}}; \tag{7}$$

where indexes *a*, *b* are relating to ISCCS using solar energy for steam methane reforming and a traditional ISCCS using solar energy for steam generation, respectively; Q_{exh}, Q_{solar} – exhaust heat and solar energy.

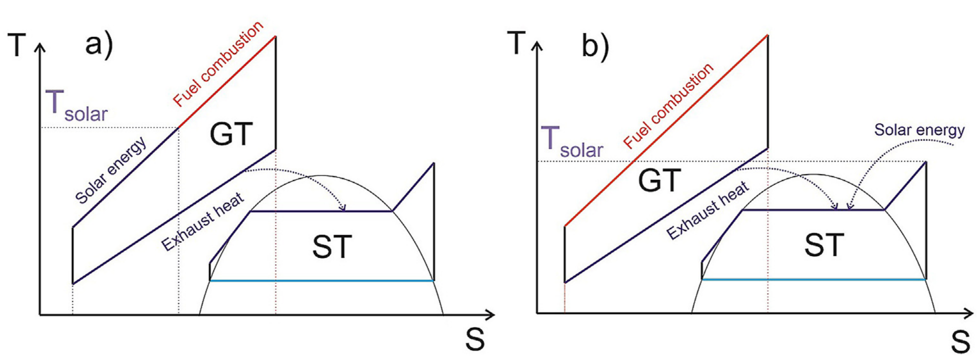

Fig. 2 – Thermodynamic cycles of ISCCS with using solar energy for steam methane reforming (a) and a conventional ISCCS with using solar energy for steam generation (b).

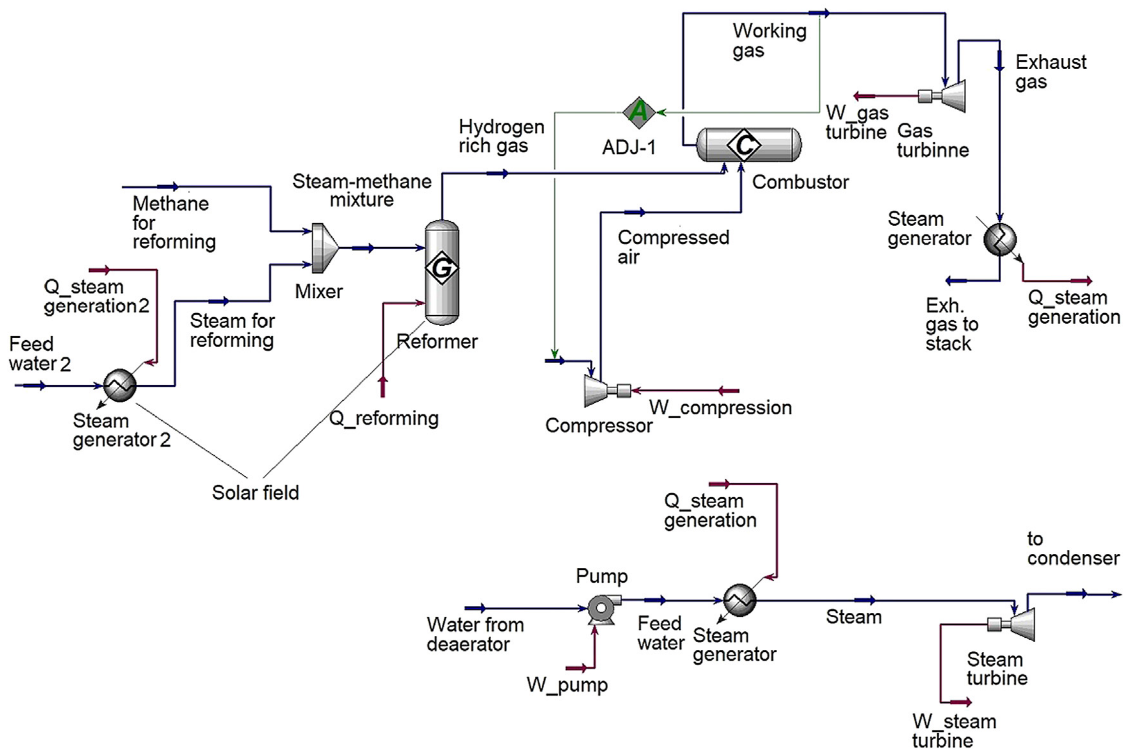

Fig. 3 – Computational scheme of ISCCS with steam methane reforming in Aspen HYSYS.

A thermodynamic analysis of ISCCS where solar energy is utilized for steam methane reforming and a traditional ISCCS where solar energy is utilized for a steam generation have been performed via Aspen HYSYS [43–45].

A computational scheme of an ISCCS with the use of solar energy for steam methane reforming from Fig. 1a in Aspen HYSYS is presented in Fig. 3. A computational scheme of a traditional ISCCS with the use of solar energy for steam generation is not presented. In the computational scheme of this system the solar energy is summed with exhaust heat and supplied for steam generation.

The thermodynamic analyses were performed for the different gas turbine inlet temperatures. Turbine inlet temperature (TIT) is controlled by an air mass flow rate. In Aspen HYSYS this temperature control can be performed via function (*Adjust*). Solar energy in the computational scheme is utilized for the generation of steam for the steam methane reforming process, heating of the reaction mixture, and for the endothermic reaction as follows:

$$Q_{solar} = Q_{st.gen} + \Delta Q_{mix} + \Delta H_{298} \quad (8)$$

where $Q_{st.gen}$, ΔQ_{mix}, ΔH_{298} – heat flows for steam generation, reaction mixture heating, an endothermic effect of the steam methane reforming process, respectively.

Thermodynamic model setup

A thermodynamic analysis was fulfilled for the different technological variables. Stream properties and the efficiency parameters for ISCCS and for the methane and air compressors are shown in Table 1.

Operational parameters were varied for the gas turbine cycle, while for the steam turbine cycle the pressure and temperature of steam are constant for all cases. The heat flows for the steam methane reforming process are significantly affected by temperature, pressure, and steam-to-methane ratio. In this study, the heat flows for reforming were set as the same for a conventional ISCCS where solar energy is used for steam generation. Put it otherwise, heat flow with solar energy for reforming in ISCCS (Fig. 1a) is equal to heat flow for steam generation in ISCCS (Fig. 1b).

Table 1 – The parameters of ICSSC with steam methane reforming.

Parameter	Value
CH_4 mass flow rate	100 kg/h
Turbine inlet temperature (TIT)	1000; 1200; 1400 °C
Compression/expansion ratio	10–20
Pressure losses	0
Turbine isentropic efficiency	90%
Compressors isentropic efficiency	85%
Steam temperature at steam turbine inlet	500 °C
Steam pressure at steam turbine inlet	100 bar
Condenser pressure	0.06 bar
Steam-to-methane ratio (mol-to-mol)	1.0; 2.0
Temperature of atmospheric air and methane	20 °C
Methane reforming temperature	500; 600; 700; 800 °C
Temperature of steam for reforming	330 °C
Pressure of all inlet streams (except air)	10 bar

The molar steam-to-methane ratio is paying an important role in the reforming process. Further, the steam-to-methane ratio is marked as β which is presenting a molar ratio of H_2O to CH_4 at a steam methane reformer inlet:

$$\beta = \frac{H_2O_{in}}{CH_{4,in}} \tag{9}$$

where H_2O_{in}, $CH_{4,in}$ is steam and methane moles in a reaction mixture at a reformer inlet.

Steam methane reforming

A main goal of the thermodynamic analysis of SMR is a calculation of heat and mass balances of a steam-methane chemical system. Initial steam and methane is reacting to produce hydrogen, carbon monoxide, carbon dioxide. In addition, as the reforming products contain unreformed methane and steam. Thus, it was taken into account that the reformate consists of H_2, CO, CO_2, H_2O, CH_4. However, various chemical reactions can be observed in the reformer, including the reactions of carbon formation. There is sufficient information in the literature that in the temperature range above 650 K, carbon formation can be avoided by an accurate organization of SMR process [39,46].

A reformate equilibrium composition was calculated via GFEM (Gibbs free energy minimization) technique. The thermodynamic analysis of SMR process was fulfilled via Aspen HYSYS. The detailed algorithm of thermodynamic analysis is published in author's previous papers [26,47].

A thermodynamic efficiency of the steam methane reforming process is determined by methane conversion. If the whole methane has reacted, then the methane conversion is unity. CH_4 conversion can be calculated by the relationship:

$$X_{CH_4} = \frac{CH_{4,in} - CH_{4,out}}{CH_{4,in}} \tag{10}$$

where $CH_{4,in}$, $CH_{4,out}$ is CH_4 mole fraction at inlet and outlet of steam methane reformer, respectively.

In addition, with the help of thermodynamic analysis, the heat flows in SMR system can be calculated. Whole solar energy is fed to produce steam for the reforming, to heat the steam-methane mixture, and to compensate an endothermic effect of SMR process.

Results and discussion

Steam methane reforming

Following the Le-Chatelier's principle a composition of synthesis gas after the steam methane reformer is affected by temperature, pressure, and H_2O-to-CH_4 ratio (β). CH_4 conversion increases when temperature and β ratio is increasing. The effect of temperature is observed due to the fact that the reforming reaction is endothermic. Contrariwise, CH_4 conversion decreases with increasing in pressure because the number of moles of reformate is higher than the numbers of moles of initial components. The effect of technological variables such as temperature, β ratio, and pressure on the equilibrium synthesis gas composition is presented in Table 2.

Fig. 4 plotted to better clarify an influence of technological variables (pressure, temperature and β ratio) on methane conversion which is showing the efficiency of whole reforming process.

Fig. 4 evidently shows the effect of technological variables on CH_4 conversion based on Le-Chatelier's principles described above. An increase in temperature and β ratio leads to increase in CH_4 conversion. On the other side, an increase in pressure leads to decrease in CH_4 conversion.

To understand the distribution of the heat flows in the steam methane reforming system, the heat balance of the solar field was compiled. Fig. 5 shows the heat flows of solar energy which is supplied for steam generation ($Q_{st.gen}$), reaction mixture heating ($Q_{preheat}$), and the endothermic effect of the reforming process (Q_{ref}) for pressure of 10–20 bar and the steam-to-methane ratio of 1.0 and 2.0. The results from Fig. 5 were obtained for the mass flow rate of methane of 100 kg/h.

Fig. 5 shows that the heat flows to compensate an endothermic effect of SMR process are strongly depending on temperature. Also, pressure and β ratio have a notable effect on Q_{ref}. The heat flow for steam generation is constant for each case because the steam temperature is constant. The heat flow for mixture heating is slightly increasing when the temperature increases.

Fig. 5 depicts that even for 800 °C and β = 2.0 the share of heat supplied for steam generation is about one-third. For the temperature range below 590–640 °C, solar energy is mainly

Table 2 – Effect of temperature, β ratio, and pressure on equilibrium synthesis gas composition.

T,°C	β	p = 1 MPa					p = 3 MPa				
		H_2	CO	CH_4	H_2O	CO_2	H_2	CO	CH_4	H_2O	CO_2
	1.0	8.19	0.03	45.85	43.91	2.02	5.31	0.01	47.29	46.06	1.32
400	2.0	8.47	0.02	29.76	59.65	2.10	5.47	0.01	31.01	62.15	1.36
	3.0	8.36	0.02	21.82	67.72	2.07	5.39	0.01	22.94	70.32	1.34
	1.0	29.12	2.11	34.35	28.73	5.7	20.32	0.94	39.33	35.03	4.38
600	2.0	30.02	1.64	20.11	41.96	6.27	20.95	0.74	24.26	49.36	4.69
	3.0	29.40	1.38	13.43	49.48	6.32	20.59	0.63	17.01	57.09	4.68
	1.0	55.98	14.35	14.80	11.63	3.23	43.70	8.87	23.68	19.48	4.27
800	2.0	54.90	11.43	5.67	22.85	5.15	43.81	7.19	12.05	31.39	5.56
	3.0	50.87	8.94	2.55	31.63	6.01	41.89	5.96	7.03	39.12	6.00
	1.0	70.29	22.67	3.49	2.97	0.57	63.67	19.49	8.39	7.15	1.30
1000	2.0	62.26	16.95	0.31	17.64	2.85	59.46	15.69	2.00	19.76	3.09
	3.0	53.85	12.52	0.09	29.47	4.07	52.69	12.06	0.69	30.43	4.13

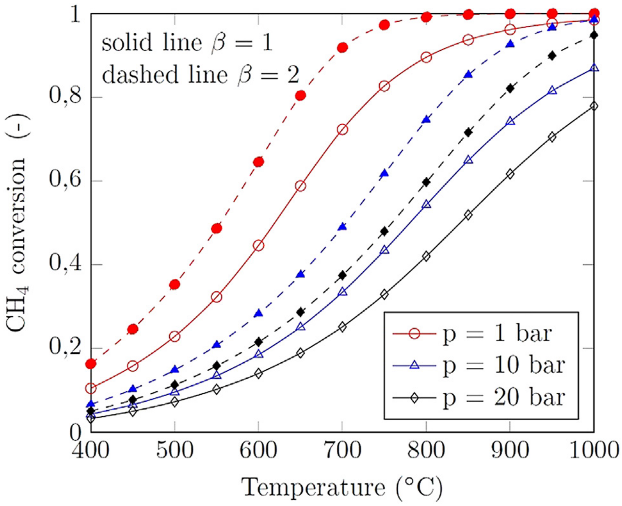

Fig. 4 – Methane conversion for various temperature, pressure, β ratio.

spent for steam generation and reaction mixture heating while the heat flow for the steam methane reforming is minimal. Q_{ref} shows the amount of solar energy that has been converted into the chemical energy of new hydrogen-rich gas. Therefore, to have a notable share of solar energy converted into chemical energy, the temperature of the process should be above 600–650 °C.

Cycle analysis

The operation parameters are also affecting the thermal efficiency of ISCCS. Fig. 6 shows the thermal efficiency for various operating parameters: temperature, pressure, and steam-to-methane ratio. The maximum efficiency is observed for a steam-to-methane ratio of 1.0 at 10 bar and 20 bar. The main reason for this is the fact that an increase in a steam-to-methane ratio leads to an increase in the heat flow for steam generation. The main part of this heat flow is supplied to evaporate liquid water (evaporation heat). In the case when β ratio is above 1.0, synthsis gas consists of a notable amount of steam (Table 2). Therefore, exhaust gas contains a notable amount of steam (4 mol of H_2O per 1 mol of CH_4 for $\beta = 2.0$, and 3 mol of H_2O per 1 mol of CH_4 for $\beta = 1.0$). In this case, the heat flow for evaporation of steam in the solar field is rejected into the atmosphere. Thus, even total power output for $\beta = 2.0$ is higher than for $\beta = 1.0$, the overall efficiency of ISCCS with steam methane reforming is smaller.

Also, Fig. 6 depicts that an increase of pressure (expansion/compression ratio) leads to an increase in the overall efficiency of both ISCCS with and without steam methane reforming. As well as an increase in turbine inlet temperature leads to increasing the overall efficiency [48]. These facts are in good agreement with the classical ideas of the efficiency of gas turbines. An increase in pressure and temperature leads to an increase in the efficiency of a gas turbine which leads to increasing the overall efficiency of ISCCS.

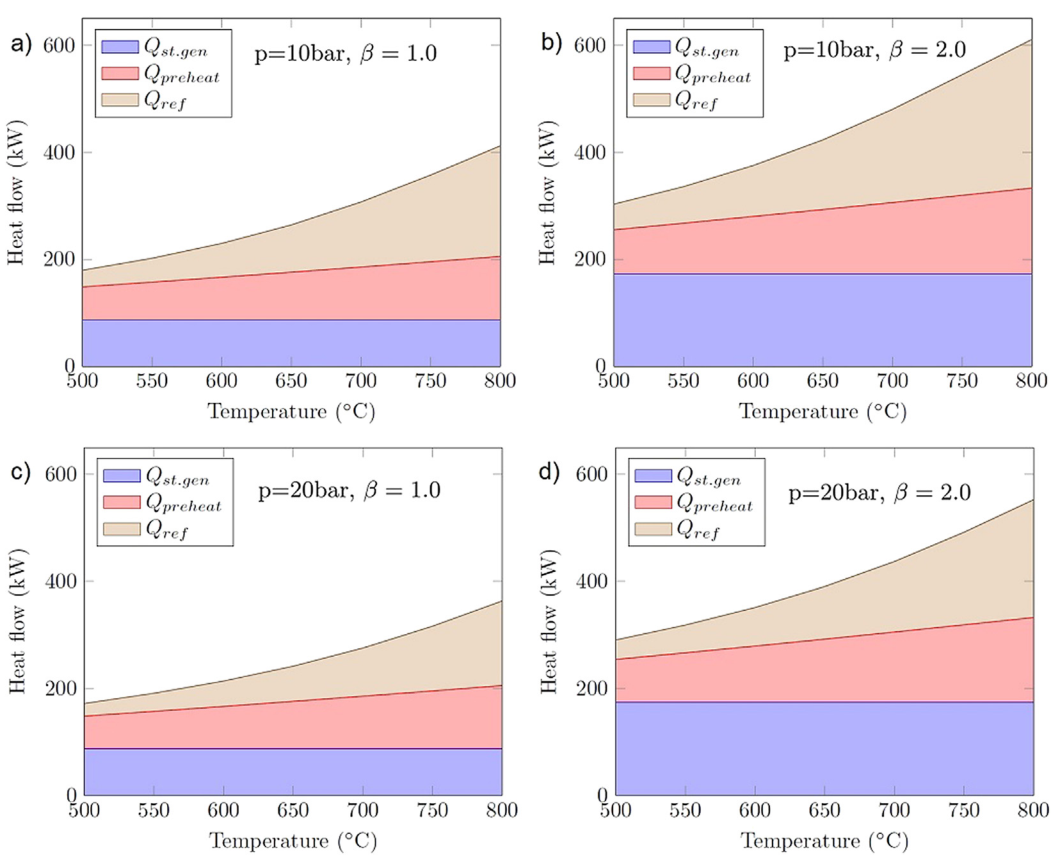

Fig. 5 – Heat balance of the steam methane reforming system.

Fig. 6 shows that an increase in the reforming temperature leads to a slight increase in the efficiency of ISCCS with steam methane reforming. However, an increase in the reforming temperature leads to a decrease in the efficiency of ISCCS without steam reforming. As was accepted above, the heat flow for the steam methane reforming process in the scheme on Fig. 1a is equal to the heat flow for the steam generation for a steam turbine in the scheme on Fig. 1b. When the reforming temperature increases, the heat flow in the solar field is increasing. Therefore, in ISCCS without steam methane reforming the power output of a steam turbine is increasing. However, the efficiency of a steam turbine cycle is lower than the efficiency of a gas turbine cycle. As a result, the overall efficiency of ISCCS without steam methane reforming is decreasing with an increase in the reforming temperature. On the overhand, an increase in the reforming temperature leads to increasing in the power output of a gas turbine. Thus, the efficiency of ISCCS with steam methane reforming is increased when the temperature increases.

To understand the distribution of the power in a gas and steam turbine, the shares of power output for a gas turbine and steam turbine cycle are presented in Fig. 7. Fig. 7 shows that the share of the power output of a gas turbine cycle in the ISCCS with steam methane reforming is higher than it in the ISCCS without steam methane reforming. Also, the share of the power output of a steam turbine cycle in the ISCCS without steam methane reforming is higher than it in the ISCCS with steam methane reforming. Therefore, the overall efficiency of the ISCCS with steam methane reforming is higher than it in the ISCCS without steam methane reforming.

Fig. 6 shows that for the reforming temperature of 500 °C the efficiency of the ISCCS with and without steam methane reforming is almost equal for $\beta = 1.0$. The efficiency for scheme on Fig. 1b is even higher for $\beta = 2.0$ at $T_{ref} = 500$ °C than it for scheme on Fig. 1a. However, the solar energy in the ISCCS with steam methane reforming is utilized in the gas turbine cycle with higher efficiency than in the steam turbine cycle. The main reason for this finding is the fact that a notable amount of solar energy is used to generate steam for the reforming process. However, this steam after reforming becomes a part of synthesis gas and after combustion becomes a part of exhaust gas. After a steam generator, exhaust gas with a high share of H_2O is released into the atmosphere. Therefore, solar energy for steam generation is lost with exhaust gas. One of the ways to avoid these energy losses is a condensation of steam from the exhaust gases and the useful use of latent heat of condensation.

Table 3 shows the overall efficiency of ISCCS for the schemes on Fig. 1b (η_b) and Fig. 1a (η_a) as well as efficiency of ISCCS with steam methane reforming when H_2O for reforming

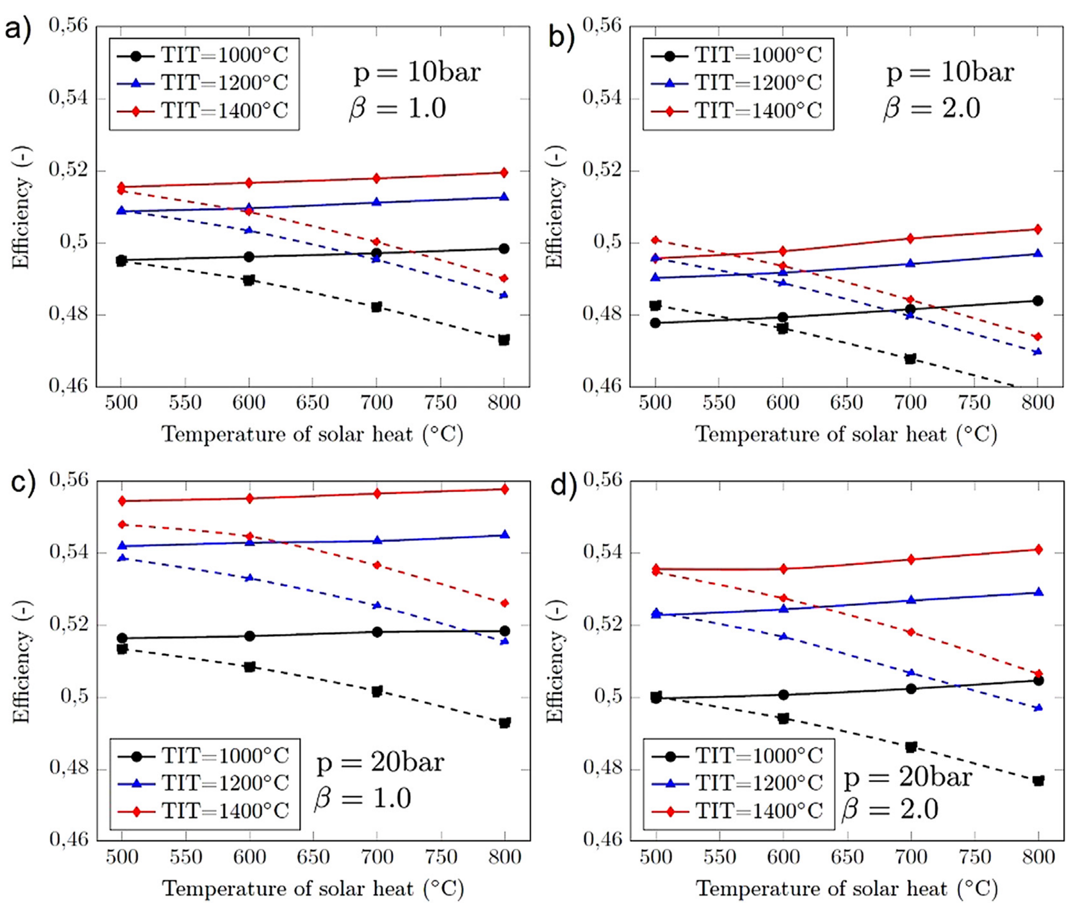

Fig. 6 – Efficiency of the integrated solar combined cycle systems for various temperature, pressure, and steam-to-methane ratio: solid lines – ISCCS with steam methane reforming (scheme Fig. 1a); dashed lines – without steam methane reforming (Fig. 1b).

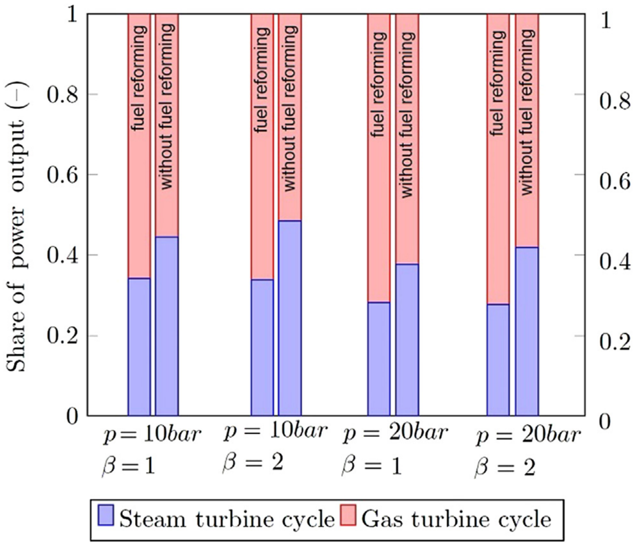

Fig. 7 – Share of power output for a gas turbine and steam turbine cycle in ISCCS with and without steam methane reforming for TIT = 1200 °C and T_{ref} = 800 °C.

is condensed from exhaust gas (η_{cond}). Based on the material balance, if a steam-to-methane ratio is 1.0, exhaust gas contains 1 additional mole of H_2O per 1 mol of CH_4. If β ratio is 2.0, exhaust gas contains 2 additional moles of H_2O. Table 3 shows the efficiency of ISCCS with steam methane reforming when 1 or 2 mol of H_2O (for β = 1.0 and β = 2.0) were condensed and condensation heat is used for steam generation. Table 3 presents that condensation of additional H_2O from exhaust gas is increasing the overall efficiency of ISCCS with steam methane reforming up to 2.7% and 5.4% for β = 1.0 and β = 2.0, respectively. Moreover, if heat losses with uncondensed H_2O are eliminated, the overall efficiency of ISCCS with steam methane reforming is higher than it is for ISCCS without steam methane reforming.

Table 3 – Efficiency of ISCCS for the schemes on Fig. 1b (η_b) and Fig. 1a (η_a) as well as efficiency of ISCCS with steam methane reforming when H_2O for reforming is condensed from exhaust gas (η_{cond}).

		p = 10 bar			p = 20 bar		
		η_b	η_a	η_{cond}	η_b	η_a	η_{cond}
TIT, °C	β	T_{ref} = 800 °C			T_{ref} = 800 °C		
1000	1.0	47.31	49.84	52.35	49.29	51.84	54.05
	2.0	45.86	48.40	52.95	47.68	50.47	55.37
1200	1.0	48.54	51.27	53.84	51.54	54.49	56.82
	2.0	46.97	49.70	54.37	49.69	52.90	58.04
1400	1.0	49.01	51.95	54.57	52.60	55.77	58.15
	2.0	47.39	50.38	55.12	50.64	54.10	59.36
		T_{ref} = 500 °C			T_{ref} = 500 °C		
1000	1.0	49.50	49.53	52.41	51.34	51.64	54.65
	2.0	48.27	47.78	53.18	50.01	49.97	55.66
1200	1.0	50.90	50.88	53.84	53.85	54.19	57.36
	2.0	49.58	49.03	54.57	52.36	52.27	58.23
1400	1.0	51.44	51.56	54.55	54.79	55.45	58.69
	2.0	50.07	49.58	55.18	53.47	53.43	59.51

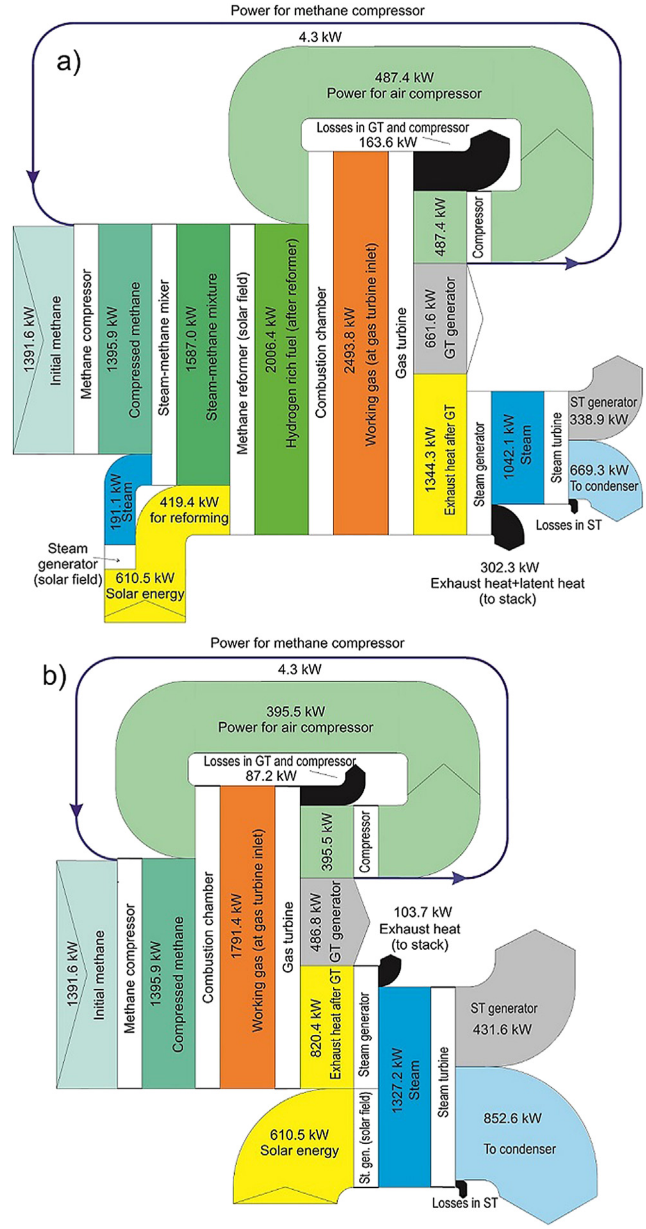

Fig. 8 – Sankey diagram for ISCCS with (Fig. 1a) and without (Fig. 1b) steam methane reforming for TIT = 1200 °C, T_{ref} = 800 °C, p = 10 bar, β = 2.0.

Fig. 8 shows the Sankey diagram for ISCCS with (Fig. 1a) and without (Fig. 1b) steam methane reforming for TIT = 1200 °C, T_{ref} = 800 °C, p = 10 bar, β = 2.0. Fig. 8 depicts that the utilization of solar energy for the steam methane reforming processes is increasing the overall power output in a gas turbine cycle. In parallel, the utilization of solar energy for the steam methane reforming process leads to increasing in exhaust heat (to stack) due to the high volume fraction of H_2O in the exhaust gases. Therefore, the investigation of an integrated solar combined cycle system with steam methane reforming with H_2O condensation is a practically interesting task for future investigations.

Moreover, Fig. 8 shows that power output of a steam turbine cycle in ISCCS with using solar energy for steam generation is increasing, because the power of the compressor and gas turbine is lower due to the reduction of heat in a gas turbine cycle.

Sankey diagram also shows the perspectives for the investigation of ISCCS with steam methane reforming when steam for the steam methane reforming process is extracted from a steam turbine.

Conclusion

In this paper, the thermodynamic analysis of ISCCS using solar energy for steam methane reforming was performed. The results of the thermodynamic analysis were compared with a conventional integrated solar combined cycle system using solar energy for steam generation for a steam turbine cycle. It was established that an increase in turbine inlet temperature and pressure in a gas turbine leads to increasing in the overall efficiency. However, an increase in β ratio above 1.0 leads to decreasing in the overall efficiency. If water that used for steam methane reforming will be condensed from the exhaust gases, the overall efficiency of ISCCS with steam methane reforming will increase up to 6.2% and 8.9% for $\beta = 1.0$ and $\beta = 2.0$, respectively, in comparison with ISCCS where solar energy is utilized for steam generation for a steam turbine. The maximum efficiency in the investigated range of the operational parameters is 55.77% (TIT = 1400 °C, T_{ref} = 800 °C, $\beta = 1.0$, p = 20 bar) for ISCCS with steam methane reforming without H_2O condensation and 59.51% (TIT = 1400 °C, T_{ref} = 800 °C, $\beta = 2.0$, p = 20 bar) with H_2O condensation.

Utilization of solar energy for steam methane reforming increases the share of power of a gas turbine cycle: two-thirds are in a gas turbine cycle, and one-third is in a steam turbine cycle. In parallel, if solar energy is used for steam generation for a steam turbine cycle the shares of power from a gas and steam turbine are almost equal.

The detailed investigation of ICSSC with steam methane reforming with H_2O condensation as well as with extraction of steam from a steam turbine for the methane reforming process is an interesting task for future investigations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the Russian Science Foundation under grant 19-19-00327.

REFERENCES

[1] Dincer Ibrahim. Renewable energy and sustainable development: a crucial review. Renew Sustain Energy Rev 2000;4(2):157–75.

[2] Balezentis Tomas, Streimikiene Dalia, Mikalauskas Ignas, Shen Zhiyang. Towards carbon free economy and electricity: the puzzle of energy costs, sustainability and security based on willingness to pay. Energy 2021;214:119081.

[3] R Timilsina Govinda, Kurdgelashvili Lado, Narbel Patrick A. Solar energy: markets, economics and policies. Renew Sustain Energy Rev 2012;16(1):449–65.

[4] Kannan Nadarajah, Vakeesan Divagar. Solar energy for future world:-a review. Renew Sustain Energy Rev 2016;62:1092–105.

[5] Shoeibi Shahin, Kargarsharifabad Hadi, Sadi Meisam, Ahmad Arabkoohsar, Mirjalily Seyed Ali Agha. A review on using thermoelectric cooling, heating, and electricity generators in solar energy applications. Sustain Energy Technol Assessments 2022;52:102105.

[6] Ahmad Arabkoohsar, Sadi Meisam. Thermodynamics, economic and environmental analyses of a hybrid waste–solar thermal power plant. J Therm Anal Calorim 2021;144(3):917–40.

[7] Ahmad Arabkoohsar, Sadi Meisam. A hybrid solar concentrating-waste incineration power plant for cost-effective and dispatchable renewable energy production. 2018 IEEE 7th international conference on power and energy (PECon), vols. 1–6. IEEE; 2018.

[8] Arabkoohsar A, Andresen GB. A smart combination of a solar assisted absorption chiller and a power productive gas expansion unit for cogeneration of power and cooling. Renew Energy 2018;115:489–500.

[9] Ritchie Hannah, Roser Max. Electricity mix. Our World in Data; 2020.

[10] Kamil Mohammed Mohammed, Omar I Awad, Rahman MM, Najafi G, Basrawi Firdaus, Abd Alla Ahmed N, Mamat Rizalman, et al. The optimum performance of the combined cycle power plant: a comprehensive review. Renew Sustain Energy Rev 2017;79:459–74.

[11] Horlock John H. Advanced gas turbine cycles: a brief review of power generation thermodynamics. 2013.

[12] Hada S, Tsukagoshi K, Masada J, Ito E. Test results of the world's first 1,600 c j-series gas turbine. Mitsubishi Heavy Industries Technical Review 2012;49(1):18.

[13] Kribus A, Zaibel R, Carey D, Akiba Segal, Jacob Karni. A solar-driven combined cycle power plant. Sol Energy 1998;62(2):121–9.

[14] Albert Boretti, Al-Zubaidy Sarim. A case study on combined cycle power plant integrated with solar energy in Trinidad and Tobago. Sustain Energy Technol Assessments 2019;32:100–10.

[15] Hosseini REZA, Soltani M, Valizadeh G. Technical and economic assessment of the integrated solar combined cycle power plants in Iran. Renew Energy 2005;30(10):1541–55.

[16] Zhu Guangdong, Neises Ty, Craig Turchi, Bedilion Robin. Thermodynamic evaluation of solar integration into a natural gas combined cycle power plant. Renew Energy 2015;74:815–24.

[17] Sheu Elysia J, Mokheimer Esmail MA, Ghoniem Ahmed F. A review of solar methane reforming systems. Int J Hydrogen Energy 2015;40(38):12929–55.

[18] Jin Jian, Wei Xin, Liu Mingkai, Yu Yuhang, Li Wenjia, Kong Hui, Hao Yong. A solar methane reforming reactor design with enhanced efficiency. Appl Energy 2018;226:797–807.

[19] He Yijian, Zheng Shupeng, Xiao Gang. Solar hybrid steam-injected gas turbine system with novel heat and water recovery. J Clean Prod 2020;276:124268.

[20] Behar Omar. A novel hybrid solar preheating gas turbine. Energy Convers Manag 2018;158:120–32.

[21] Mokheimer Esmail MA, Dabwan Yousef N, Habib Mohamed A. Optimal integration of solar energy with fossil fuel gas turbine cogeneration plants using three different csp technologies in Saudi Arabia. Appl Energy 2017;185:1268–80.
[22] Lu Yuzheng, Zhu Bin, Wang Jun, Zhang Yaoming, Li Junjiao. Hybrid power generation system of solar energy and fuel cells. Int J Energy Res 2016;40(6):717–25.
[23] Pashchenko Dmitry. Natural gas reforming in thermochemical waste-heat recuperation systems: a review. Energy; 2022, 123854.
[24] Yang Jiamin, Xiao Gang, Ghavami Mohsen, Al-Zaili Jafar, Yang Tianfeng, Sayma Abdulnaser, Dong Ni. Thermodynamic modelling and real-time control strategies of solar micro gas turbine system with thermochemical energy storage. J Clean Prod 2021;304:127010.
[25] Pashchenko Dmitry. Industrial furnaces with thermochemical waste-heat recuperation by coal gasification. Energy 2021;221:119864.
[26] Pashchenko Dmitry, Mustafin Ravil, Karpilov Igor. Efficiency of chemically recuperated gas turbine fired with methane: effect of operating parameters. Applied Thermal Engineering; 2022, 118578.
[27] Carapellucci Roberto, Milazzo Adriano. Thermodynamic optimization of a reheat chemically recuperated gas turbine. Energy Convers Manag 2005;46(18–19):2936–53.
[28] Sadeghi Mohsen, Chitsaz Ata, Marivani Parisa, Yari Mortaza, Mahmoudi SMS. Effects of thermophysical and thermochemical recuperation on the performance of combined gas turbine and organic rankine cycle power generation system: thermoeconomic comparison and multi-objective optimization. Energy 2020;210:118551.
[29] Liu Xiao, Zheng Hongtao, Liu Qian. Research of methane reforming and combustion characteristics in chemically recuperated gas turbine. Ind Eng Chem Res 2014;53(5):1940–6.
[30] Bulfin Brendan, Ackermann Simon, Furler Philipp, Steinfeld Aldo. Thermodynamic comparison of solar methane reforming via catalytic and redox cycle routes. Sol Energy 2021;215:169–78.
[31] Ma Jing, Jiang Bo, Lin Li, Yu Kewei, Zhang Qian, Zheng Lv, Tang Dawei. A high temperature tubular reactor with hybrid concentrated solar and electric heat supply for steam methane reforming. Chem Eng J 2022;428:132073.
[32] Said Syed AM, Waseeuddin Mohammed, Simakov David SA. A review on solar reforming systems. Renew Sustain Energy Rev 2016;59:149–59.
[33] Bianchini Augusto, Pellegrini Marco, Saccani Cesare. Solar steam reforming of natural gas integrated with a gas turbine power plant. Sol Energy 2013;96:46–55.
[34] Liu Taixiu, Zhang Bai, Zheng Zhimei, Liu Qibin, Lei Jing, Sui Jun, Jin Hongguang. 100 kwe power generation pilot plant with a solar thermochemical process: design, modeling, construction, and testing. Appl Energy 2019;251:113217.
[35] Wang Fu, Li Hailong, Zhao Jun, Deng Shuai, Yan Jinyue. Technical and economic analysis of integrating low-medium temperature solar energy into power plant. Energy Convers Manag 2016;112:459–69.
[36] Ma Jing, Jiang Bo, Wang Si-ma, Yu Kewei, Zheng Lv, Gao Yuming, Yuan Dazhong, Tang Dawei. Deciphering high-efficiency solar-thermochemical energy conversion process of heat pipe reactor for steam methane reforming. Fuel 2022;326:124972.
[37] Ishaq H, Dincer I. Analysis and optimization for energy, cost and carbon emission of a solar driven steam-autothermal hybrid methane reforming for hydrogen, ammonia and power production. J Clean Prod 2019;234:242–57.
[38] Xu Jianguo, Froment Gilbert F. Methane steam reforming, methanation and water-gas shift: I. Intrinsic kinetics. AIChE J 1989;35(1):88–96.
[39] Pashchenko Dmitry, Makarov Ivan. Carbon deposition in steam methane reforming over a ni-based catalyst: experimental and thermodynamic analysis. Energy 2021;222:119993.
[40] Tabrizi Farshad Farshchi, Mousavi Seyed Amir Hossein Seyed, Atashi Hossein. Thermodynamic analysis of steam reforming of methane with statistical approaches. Energy Convers Manag 2015;103:1065–77.
[41] Pashchenko Dmitry. Thermodynamic equilibrium analysis of steam methane reforming based on a conjugate solution of material balance and law action mass equations with the detailed energy balance. Int J Energy Res 2020;44(1):438–47.
[42] Pashchenko Dmitry. Low-grade heat utilization in the methanol-fired gas turbines through a thermochemical fuel transformation. Therm Sci Eng Prog 2022;36:101537.
[43] Liu Zuming, Karimi Iftekhar A. Simulating combined cycle gas turbine power plants in aspen hysys. Energy Convers Manag 2018;171:1213–25.
[44] Canepa Roberto, Wang Meihong. Techno-economic analysis of a co2 capture plant integrated with a commercial scale combined cycle gas turbine (ccgt) power plant. Appl Therm Eng 2015;74:10–9.
[45] Harvey Simon, et al. Analysis of a reheat gas turbine cycle with chemical recuperation using aspen. Energy Convers Manag 1997;38(15–17):1671–9.
[46] Trimm David L. Coke formation and minimisation during steam reforming reactions. Catal Today 1997;37(3):233–8.
[47] Pashchenko Dmitry, Mustafin Ravil, Karpilov Igor. Ammonia-fired chemically recuperated gas turbine: thermodynamic analysis of cycle and recuperation system. Energy 2022;252:124081.
[48] Kotowicz Janusz, Job Marcin, Brzkeczek Mateusz, Nawrat Krzysztof, Mkedrych Janusz. The methodology of the gas turbine efficiency calculation. Arch Therm 2016;(4).